FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

# JEE (Main), 2015

# HALF COURSE TEST – VII

**Time Allotted: 3 Hours** **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## INSTRUCTIONS

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

**(i)** **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

| | |
|---|---|
| **Name of the Candidate** | |
| **Enrolment No.** | |

## Useful Data

### PHYSICS

| | |
|---|---|
| Acceleration due to gravity | $g = 10\ m/s^2$ |
| Planck constant | $h = 6.6 \times 10^{-34}$ J-s |
| Charge of electron | $e = 1.6 \times 10^{-19}$ C |
| Mass of electron | $m_e = 9.1 \times 10^{-31}$ kg |
| Permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$ |
| Density of water | $\rho_{water} = 10^3\ kg/m^3$ |
| Atmospheric pressure | $P_a = 10^5\ N/m^2$ |
| Gas constant | $R = 8.314\ J\ K^{-1}\ mol^{-1}$ |

### CHEMISTRY

| | | | |
|---|---|---|---|
| Gas Constant | R | = | $8.314\ J\ K^{-1}\ mol^{-1}$ |
| | | = | $0.0821$ Lit atm $K^{-1}\ mol^{-1}$ |
| | | = | $1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$ |
| Avogadro's Number | $N_a$ | = | $6.023 \times 10^{23}$ |
| Planck's constant | h | = | $6.625 \times 10^{-34}$ J·s |
| | | = | $6.625 \times 10^{-27}$ erg·s |
| 1 Faraday | | = | 96500 coulomb |
| 1 calorie | | = | 4.2 joule |
| 1 amu | | = | $1.66 \times 10^{-27}$ kg |
| 1 eV | | = | $1.6 \times 10^{-19}$ J |

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

# Physics

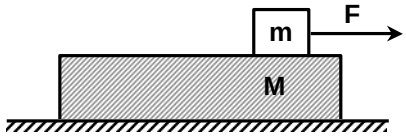
PART – I

## SECTION – A

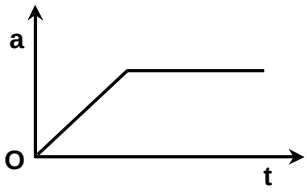
### Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

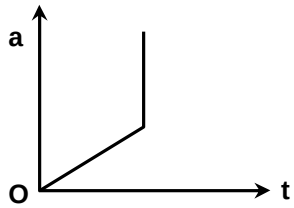
1. A block of mass m is placed on the plank of mass M as shown in the figure. Coefficient of friction between the surfaces of m and M is $\mu$ and there is no friction between ground and M. A horizontal force F = kt is applied on the block, where k is a constant and t is the time. Which of the following graphs shows the acceleration of the plank?



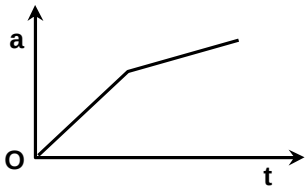

(A)



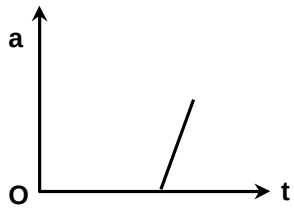

(B)



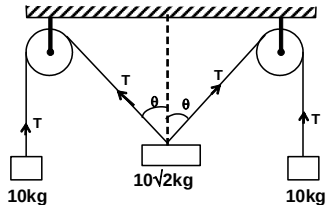

(C)




(D)




2. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle $\theta$ should be




(A) 0° (B) 30°

(C) 45° (D) 60°

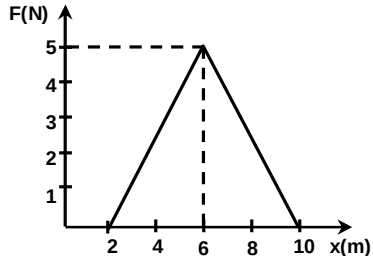
3. A body of mass m slides down on an incline plane. It reaches at the bottom with a velocity v. If the same mass were in the form of the ring which rolls down this inclined, the velocity of the ring at the bottom's

(A) $\sqrt{2}v$ (B) $\frac{1}{\sqrt{2}}v$

(C) v (D) 2v

***Space for rough work***

4. A force shown in the F-x graph is applied to a 2 kg block horizontal as shown in the figure. The change in kinetic energy is
(A) 15 J
(B) 20 J
(C) 25 J
(D) 30 J




5. Two masses of 4 kg and 1 kg are moving in opposite direction with equal kinetic energies of 8 J. If the collision between the masses are perfectly inelastic the percentage loss of kinetic energy of the masses is
(A) 60% (B) 70%
(C) 80% (D) 90%

6. A particle of mass m and radius is moving in a horizontal circle of radius r with a centripetal force $F=-\frac{k}{r^2}\hat{r}$, where k = 1 unit. The total energy of the particle is
(A) $-\frac{k}{r}$ (B) $\frac{1.5k}{r}$
(C) $-\frac{k}{2r}$ (D) $-\frac{3k}{2r}$

7. A particle moves in the x-y plane under the action of a force $\vec{F}$ at any time t its linear momentum is given by $P_x = 2\cos(kt)$ and $P_y = 2\cos(kt)$
The angle between $\vec{F}$ and $\vec{P}$ at time t is
(A) 0° (B) 30°
(C) 90° (D) 180°

8. An ideal spring having force constant k is suspended from the rigid support and a block of mass M is attached to its lower end. The mass is released from the natural length of the spring. Then the maximum extension in the spring is
(A) $\frac{4Mg}{k}$ (B) $\frac{2Mg}{k}$
(C) $\frac{Mg}{k}$ (D) $\frac{Mg}{2k}$

***Space for rough work***

9. A particle of mass m is moving with a uniform speed v in x-y plane along a straight line, y = x+k, the angular momentum of the particle about the origin is

(A) $\frac{mvk}{\sqrt{2}}$ (B) $\frac{mvy}{\sqrt{2}}$

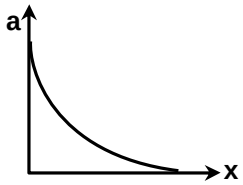
(C) $\frac{mvx}{\sqrt{2}}$ (D) mvk

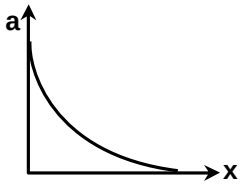
10. A disc and a ring having same mass M and radius R rotating about its axis which is normal to their plane. If both have the same angular momentum, then the ratio of acceleration of the particles at the circumference of the disc to the ring is

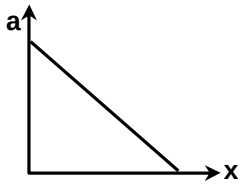
(A) 1 : 2 (B) 2 : 1

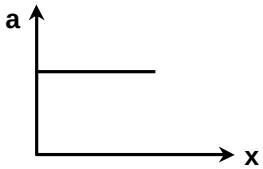
(C) 1 : 4 (D) 4 : 1

11. A spring-block system can oscillates on a smooth horizontal surface. If the spring is compressed and is released from rest. The graph between the acceleration of the block and the distance x travelled by it is given by

(A) 


(B) 


(C) 


(D) 


12. A hollow sphere is placed on a rough horizontal surface. A sharp impulse is given to the sphere horizontally towards its centre

(A) A frictional force acts on the sphere throughout the motion.
(B) kinetic energy remains constant throughout the motion.
(C) kinetic energy of translation changes into rotation but total kinetic energy decreases initially and finally becomes constant.
(D) Work is done by the frictional force throughout the motion.

***Space for rough work***

13. The initial velocity of a particle is $\vec{u} = \left(2\hat{i} + 3\hat{j}\right)$m/s. A constant force $\vec{F} = \left(4\hat{i} + 3\hat{j}\right)$N acts on the particle. Hence

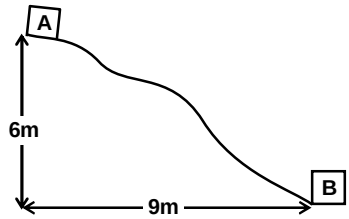
(A) Its velocity is constant.
(B) Its path is parabolic.
(C) its path is straight line.
(D) it moves on a circular path.

14. A particle moves along a curve $y = \frac{x^2}{2}$ and $x = \frac{t^2}{2}$, where x and y are measured in metre and t in seconds. At t = 2 sec. The velocity of the particle is

(A) $\left(2\hat{i} - 4\hat{j}\right)$m/s
(B) $\left(2\hat{i} + 4\hat{j}\right)$m/s
(C) $\left(4\hat{i} + 2\hat{j}\right)$m/s
(D) $\left(4\hat{i} - 2\hat{j}\right)$m/s

15. The 300 N crate slides down the curved path from A to B in the vertical plane as shown in the figure. If the crate has a velocity of 1.2 m/s down the inclined at point A and 8 m/s at B. Find the work done against friction during the motion

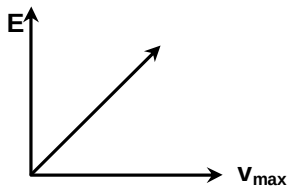
(A) 860 N m
(B) 987 N m
(C) 1200 N m
(D) 500 N m

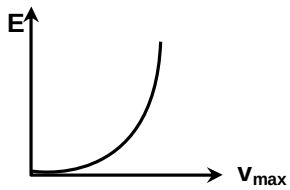


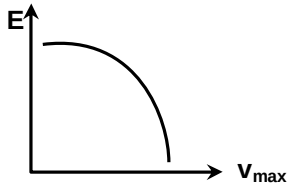

16. A block B of specific gravity 2 and another block C of specific gravity 0.5. Both are joined together and they are floating inside water such that they are completely dipped inside water, the ratio of the masses of the blocks B and C is

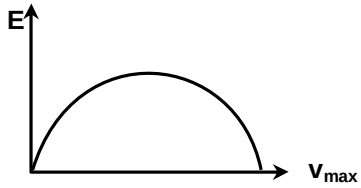
(A) 2 : 1
(B) 3 : 2
(C) 5 : 3
(D) 2 : 3

17. A source of sound emits sound waves in a uniform medium. If energy density is E and maximum speed of particles of the medium is $v_{max}$ then the graph between E and $v_{max}$ is best represented by

(A)



(B)



(C)



(D)


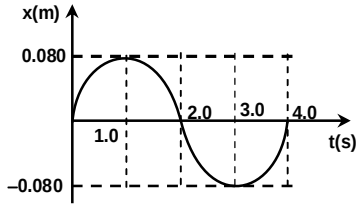

***Space for rough work***

18. The potential energy of a harmonic oscillation of mass 2 kg in its resting position is 5 joules. Its total energy is 9 joules and its amplitude is 1 cm. It is time period will be

(A) $3.14\times10^{2}$ sec (B) $3.14\times10^{-2}$ sec

(C) $10^{3}$ sec (D) data are not sufficient.

19. An object of mass 0.8 kg is attached to one end of a spring and the system is set into simple harmonic motion. The displacement x of the object as a function of time is shown in the figure with the aid of the data the magnitude of the objects acceleration at t = 1.0 is




(A) zero. (B) 1.57 $m/s^2$

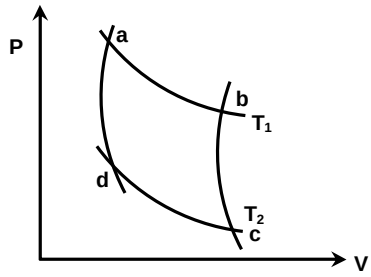
(C) 0.197 $m/s^2$ (D) 0.157 $m/s^2$

20. An air bubble 0.8 mm in diameter having density 1.293 $kg/m^3$ rise in a liquid of viscosity 0.15N-$s/m^2$ and specific gravity 0.9 with terminal velocity.

(A) 0.21 cm/s (B) 0.42 cm/s

(C) 0.11 cm/s (D) 0.183 cm/s

21. Two different adiabatic paths for the same gas intersects two isothermal $T_1$ and $T_2$ as shown in the P-V diagram. Then $V_a/V_d$ is equal to




(A) $\frac{V_C}{V_b}$ (B) $V_C V_b$

(C) $\frac{V_b}{V_c}$ (D) $\frac{V_C V_b}{V_C + V_b}$

22. The potential energy of a particle oscillating along x-axis is given as U = 20+ $(x-2)^2$ where U is in joules and x in meters. Total mechanical energy of the particle is 36 J. Maximum kinetic energy of the particle is

(A) 24 J (B) 36 J

(C) 16 J (D) 20 J

23. Two spheres of the same material and same radius r are touching each other gravitational force between them is directly proportional to

(A) $r^2$ (B) $r^3$

(C) $r^4$ (D) r

***Space for rough work***

24. Force of attraction on a particle of mass m placed at the centre of a semicircular wire of length L and mass M is

(A) $\frac{GMm}{r^2}$ (B) $\frac{GMm}{2r^2}$

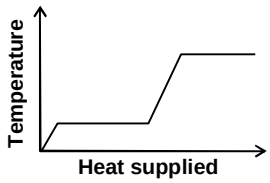
(C) zero (D) $\frac{2GMm\pi}{L^2}$

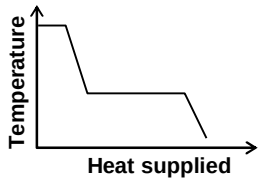
25. Starting with the same initial conditions if an ideal gas expands from volume $V_1$ to $V_2$ in three different ways. The work done by the gas is $W_1$ in the process is purely isothermal, $W_2$ is purely isobaric and $W_3$ is purely adiabatic, then

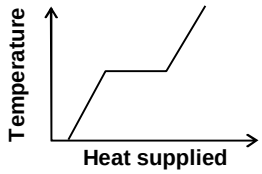
(A) $W_2 > W_1 > W_3$ (B) $W_2 > W_3 > W_1$

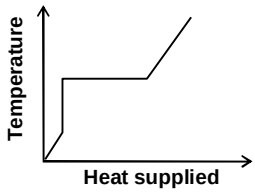
(C) $W_1 > W_2 > W_3$ (D) $W_1 > W_3 > W_2$

26. A block of ice at $-10°C$ is slowly heated and converted to steam at 100°C. Which of the following curve represents the phenomenon qualitatively?

(A)



(B)



(C)



(D)



***Space for rough work***

27. A hollow sphere of radius R is made of metal whose specific gravity is $\rho$. The sphere will float in water if thickness of wall of the sphere is (thickness of the wall << R)

(A) $> \frac{R}{3\rho}$ (B) $= \frac{2R}{3\rho}$

(C) $< \frac{R}{3\rho}$ (D) $< \frac{4R}{3\rho}$

28. A tank containing a liquid, has two similar holes of area a on the opposite sides. The difference in height of two holes is y. If the tank has an open top and liquid jets out of the two holes, the net horizontal force experienced by the tank is given by

(A) $a\rho gy$ (B) $2a\rho gy$

(C) $\frac{1}{2}a\rho gy$ (D) $\frac{2}{3}a\rho gy$

29. When the temperature of a body increases from T to T + $\Delta$T, its moment of inertia increases from I to I + $\Delta$I. The coefficient of linear expansion of the body is $\alpha$. The ratio $\Delta I/I$ is equal to

(A) $\frac{\Delta T}{T}$ (B) $\frac{2\Delta T}{T}$

(C) $\alpha T$ (D) $2\alpha\Delta T$

30. Water rises in a capillary tube to a height of 2.0 cm. In another capillary tube whose radius is one third of it, water will rise to

(A) 2 cm (B) 6 cm

(C) $\frac{2}{3}$cm (D) none of the above

***Space for rough work***

# Chemistry

PART – II

## SECTION – A

### Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Find the orbital angular momentum of an electron having quantum numbers:

   $n = 3,\ \ell = 2, m = 0, m_s = -\frac{1}{2}$

   (A) $\frac{3h}{2\pi}$ (B) $3\hbar$

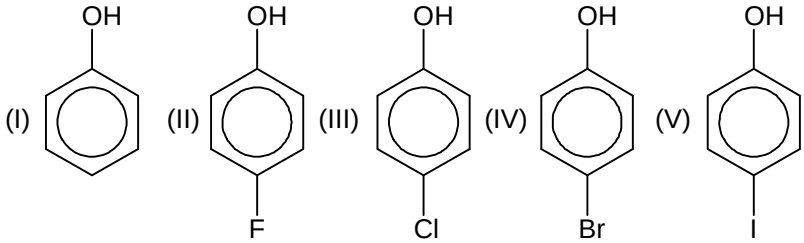
   (C) $\sqrt{6}\hbar$ (D) $\sqrt{15}$ B.M.

2. Oxidation number of carbon in carbon suboxide is

   (A) $+\frac{2}{3}$ (B) $+\frac{4}{3}$

   (C) +4 (D) $-\frac{4}{3}$

3. Arrange the following according to their increasing order of acidic strength:

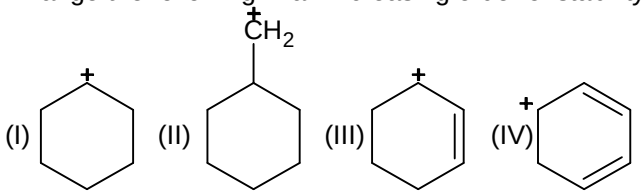
   (A) I < II < III < IV < V (B) III < I < II < IV < V

   (C) I < II < V < IV < III (D) V < IV < III < II < I

***Space for rough work***

4. 10 moles of $O_2$ gas are contained in a cylinder of 5 litre capacity. If the cylinder leakage into atmosphere (pressure is 1 atm), then the work done by the gas will be
(A) $-1.59$ kJ (B) $-2.55$ kJ
(C) $-3.27$ kJ (D) $-4.89$ kJ

5. Arrange the following in an increasing order of stability:




(A) I < II < III < IV (B) IV < III < II < I
(C) II < I < III < IV (D) III < I < IV < II

6. 90 gram of vicinal tetra bromopropane was reacted with $NaNH_2$ to form alkyne which then passes through ammonical $AgNO_3$ solution. The weight of precipitate formed will be
(A) 10.25 gm (B) 36.75 gm
(C) 46.50 gm (D) 54.28 gm

7. The volume of 0.1 M HCl required to react completely with 1 gm mixture of $Na_2CO_3$ and $NaHCO_3$ containing equimolar amount of two is
(A) 106 ml (B) 128 ml
(C) 156 ml (D) 212 ml

8. The volume of $Cl_2$ at STP react with 1 gm Fe to produce $FeCl_3$ will be
(A) 0.6 litre (B) 1.2 litre
(C) 1.8 litre (D) 2.2 litre

9. An electron whose total energy is $-2.42 \times 10^{-19}$ J given up a photon as its energy fall to $-21.76 \times 10^{-19}$ J. What is the wavelength of the emitted light?
(A) 806 $\mathring{A}$ (B) 1027 $\mathring{A}$
(C) 2812 $\mathring{A}$ (D) 5408 $\mathring{A}$

***Space for rough work***

10. What volume of $N_2$ at 300°C and 2.5 atm contains as many molecules as 2.5 L $O_2$ at STP?
(A) 1.6 L (B) 2.1 L
(C) 2.5 litre (D) 3.2 litre

11. The solubility of $Al(OH)_3$ at pH = 4, $K_{sp}$ of $Al(OH)_3 = 5 \times 10^{-33}$
(A) $5 \times 10^{-3}$ (B) $10 \times 10^{-4}$
(C) $1.53 \times 10^{-8}$ (D) $2.6 \times 10^{-3}$

12. The dipole moment of NaCl molecule is 8.5 D and its bond length is 2.36 $\mathring{A}$. The percentage ionic character in this molecule will be
(A) 50.25 % (B) 60.30 %
(C) 75.03% (D) 100%

13. The solubility of $Cd(CN)_2$ in 0.050 M NaCN solution will be what? If the $K_{sp}$ of $Cd(CN)_2$ is $1.0 \times 10^{-8}$ m
(A) $1 \times 10^{-6}$ (B) $1 \times 10^{-8}$
(C) $2 \times 10^{-8}$ (D) $4 \times 10^{-6}$

14. Calculate free energy change for the following reaction at 298 K:
$2NO(g) + Br_2(\ell) \longrightarrow 2NOBr(g)$
Given the partial pressure of NO is 0.1 atm and the partial pressure of NOBr is 2.0 atm and $\Delta G^0_{f\,NOBr} = 82.4\ kJmol^{-1}$, $\Delta G^0_{NO} = 86.55\ kJmol^{-1}$
(A) 6.7 kJ (B) –8.3 kJ
(C) –1.4 kJ (D) +8.3 kJ

15. What will be the activation energy of the reaction for which the rate of reaction triples with the temperature change by 15°C, from 298 K to 313 K?
(A) 36 kJ $mol^{-1}$ (B) 40 kJ mole
(C) 48 kJmole (D) 57 kJ/$mole^{-1}$

16. A 200 ml solution of NaOH requires 100 ml of 0.44 M aqueous HCl for complete neutralization. The temperature of the mixture increased by 1.96 K. Calculate the enthalpy change for the neutralization of one mole of hydrogen ions. Given specific heat of water = 4.184 J/g K.
(A) –56 kJ (B) 40 kJ
(C) –90 kJ (D) –110 kJ

***Space for rough work***

17. For the equilibrium:

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

If the equilibrium concentration of $PCl_5(g)$ is doubled, the concentration of $Cl_2(g)$ will be

(A) 1/2 of original value
(B) 1/4 of original value
(C) twice of original value
(D) none of these

18. Which of the following is correct for ClNO?

(A) shape is V – shaped and Cl – N – O bond angle is slightly less than 120°.
(B) shape is T – shaped and Cl – N – O bond angle is greater than 120°.
(C) shaped is linear and Cl – N – O bond angle is less than 120°
(D) none of these

19. Which of the following statements is correct for compound $HgCl_2$, $SnCl_2$ and $F_2O$ have?

(A) all have linear shape
(B) $HgCl_2$ and $SnCl_2$ are linear and $HgCl_2$ is V – shaped
(C) $HgCl_2$ is linear while $SnCl_2$ and $F_2O$ are V – shaped
(D) none of these

20. What is the molar mass of a gas if equal volume of oxygen gas and unknown gas take 3.25 and 8.41 minutes to effuse through a membrane respectively?

(A) 112 g/mole
(B) 160 g/mole
(C) 214 g/mole
(D) 306 g/mole

21. A 3.11 L container has 4.33 atm pressure at 27°C. If 2.11 L of gas at 2.55 atm is added to this container at same temperature. What will be the new pressure in the container?

(A) 4 atm
(B) 4.6 atm
(C) 5.2 atm
(D) 6.06 atm

22. What is the wavelength of light emitted by electron in $C^{5+}$ ion during transition from n = 3 to n = 1?

(A) 2.92 nm
(B) 3.6 nm
(C) 8.7 nm
(D) 9.8 nm

23. Many gases when allowed to expand suddenly, there is a drop in temperature, but there is a pair of gas which do not obey this law are

(A) $H_2$, $CO_2$
(B) $H_2$, He
(C) $Cl_2$, $O_2$
(D) $O_2$, He

***Space for rough work***

24. Calculate the weight of methane required to produce 356.4 kJ of heat on combustion. Given the heat of combustion of methane is $-890.3$ kJ
(A) 16 g (B) 6.4 g
(C) 12.8 g (D) 30.6 g

25. Molecular nitrogen can be converted into NO by the following reaction sequence,
$N_2 + 3H_2 \longrightarrow 2NH_3$
$4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O$
Calculate the mass of NO formed from 100 g of $N_2$ with excess of $H_2$ and $O_2$?
(A) 100 gm (B) 150.60 gm
(C) 214.28 gm (D) 256.32 gm

26. A compound X having molecular formula $C_4H_8$ gives Y ($C_4H_{10}O$) when reacted with $H_2O/H_2SO_4$, Y can not be resolved in optical isomer. Compound X will be
(A) $H_3C-CH=CH-CH_3$ (B) $H_3C-CH_2-CH=CH_2$
(C) $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2$ (D) None of these

27. Potassium salt of monocarboxylic acid containing n carbon atom undergo Kolbe's electrolysis then the alkane formed will contain number of carbon atom equal to
(A) 2n (B) 2n+2
(C) $2n-1$ (D) 2n – 2

28. Neopentyl bromide with alcoholic KOH gives the major product
(A) But – 2 – ene (B) 2 - methyl but – 2 – ene
(C) 2 – methyl but – 1 – ene (D) 2, 2 – dimethyl but – 1 – ene

29. Which of the following equilibrium will not be affected by adding inert gas at constant volume?
(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ (B) $3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$
(C) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (D) all the above

30. Which of the following statement is true about boron compounds?
(A) $BF_3 \Rightarrow$ least acidic among the boron halides
(B) $BF_3 \Rightarrow$ does not hydrolyses
(C) $B_2H_6 \Rightarrow$ does not have banana bond
(D) all are incorrect

***Space for rough work***

# Mathematics PART – III

## SECTION – A

### Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. A relation R : A $\to$ A be defined on set A which is union of disjoint sets $A_1A_2A_3....A_n$ having equal number of elements such that $\forall$ a $\in A_i$ and b $\in A_j$ (a, b) $\in$ R if i = j and (a, b) $\notin$ R, if i $\neq$ j
   (A) if n(A) = 15 max value of n(R) is 45 (B) R is an equivalence relation
   (C) $n(R) = [n(A)]^2$ (D) none of these

2. The value of a for which $2x^2 - 2(2a + 1)x + a(a + 1) = 0$ may have one root less that a and other root greater then 'a' are given by
   (A) 1 > a > 0 (B) –1 < a < 0
   (C) a $\geq$ 0 (D) a > 0 or a < –1

3. If $\sin^6\theta + \cos^6\theta + k\cos^2 2\theta = 1$, then k is equal to
   (A) $\frac{1}{2}\tan^2 2\theta$ (B) $\frac{1}{4}\tan^2 2\theta$
   (C) $4\cot^2 2\theta$ (D) $\frac{3}{4}\tan^2 2\theta$

4. If $\frac{\tan x}{1} = \frac{\tan y}{2} = \frac{\tan x}{3}$ ($\neq$ 0) and x + y + z = $\pi$ then
   (A) maximum value of tan x + tan y + tan z is –6 (B) minimum value of tan x + tan y + tan z is –6
   (C) tan x tan y tan z = 1 (D) tan x + tan y + tan z = 0 $\forall$ x, y, z $\in$ R

5. If the equation $ax^2 + 25x - 3c = 0$ has non real roots and $\frac{3c}{4}$ < (a + b), then c is always
   (A) negative (B) 0
   (C) non-negative (D) zero

***Space for rough work***

6. Which of the following function is surjective but not injective

(A) $f : R \to R$ $f(x) = x^4 + 2x^3 - x^2 + 1$ (B) $f : R \to R$ $f(x) = x^3 + x + 1$

(C) $f : R \to R^+$ $f(x) = \sqrt{1 + x^2}$ (D) $f : R \to R$ $f(x) = x^3 + 2x^2 - x + 1$

7. If 5a + 4b + 20c = t, then the value of t for which the line ax + by + c – 1 = 0 always passes through a fixed point is

(A) 0 (B) 20

(C) 30 (D) none of these

8. The area bounded by the circles $x^2 + y^2 = r^2$, r = 1, 2 and the rays given by $2x^2 - 3xy - 2y^2 = 0$, y > 0 is

(A) $\frac{\pi}{4}$ sq. units (B) $\frac{\pi}{2}$ sq. units

(C) $\frac{3\pi}{4}$ sq. units (D) $\pi$ sq. unit

9. The equation of the circle touching the lines |y| = x at a distance $\sqrt{2}$ units from the origin is

(A) $x^2 + y^2 - 4x + 2 = 0$ (B) $x^2 + y^2 + 4x - 2 = 0$

(C) $x^2 + y^2 + 4x + 2 = 0$ (D) none of these

10. If $\alpha$, $\beta$, $\gamma$ are the parameter of points A, B, C on the circle $x^2 + (y - 2)^2 = a^2$ and if the $\Delta ABC$ be equilateral, then

(A) $\sum \cos\alpha = 0$ (B) $\sum \sin\alpha = 0$

(C) $\sum \tan\alpha = 0$ (D) $\sum \cot\alpha = 0$

11. The orthocentre of the triangle formed by the pair of lines $2x^2 - xy - y^2 + x + 2y - 1 = 0$ and the line x + y + 1 = 0 is

(A) (–1, 0) (B) (0, 1)

(C) (–1, 1) (D) none of these

12. The equation $\sqrt{(x-2)^2 + y^2} + \sqrt{(x+2)^2 + y^2} = 4$ represents

(A) pair of lines (B) a parabola

(C) a hyperbola (D) none of these

***Space for rough work***

13. The pole of the line $\ell x + my + n = 0$ with respect to the parabola $y^2 = 4ax$

(A) $\left(-\frac{h}{\ell}, \frac{2am}{\ell}\right)$ (B) $\left(\frac{h}{\ell}, -\frac{2am}{\ell}\right)$

(C) $\left(-\frac{h}{\ell}, \frac{2am}{\ell}\right)$ (D) $\left(-\frac{h}{\ell}, -\frac{2am}{\ell}\right)$

14. If $\frac{x}{a}+\frac{y}{b}=\sqrt{2}$ touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ then the eccentric angle $\theta$ of the point of contact is equal to

(A) 0 (B) 90º

(C) 45º (D) 60º

15. If e, e′ be the eccentricities of two conics S and S′ and if $e^2 + e'^2 = 3$ then both S ands S′ can be

(A) hyperbolas (B) ellipse

(C) parabolas (D) none of these

16. Base angle of a triangle are $22\frac{1}{2}^o$ and $112\frac{1}{2}^o$ then find the height of triangle

(A) equal to base (B) one third of base

(C) half of base (D) one forth of base

17. If in a triangle $\Delta ABC$ a, b, c are roots of the equation $x^3 - 11x^2 + 38x - 40 = 0$ then $\sum\frac{\cot A}{a}$ is equal to

(A) $\frac{3}{4}$ (B) 1

(C) $\frac{9}{16}$ (D) none of these

18. Two circles with radii a and b touch each other externally such that $\theta$ is the angle between their direct common tangents ($a > b \geq 2$), then

(A) $\theta = 2\cos^{-1}\left(\frac{a-b}{a+b}\right)$ (B) $\theta = 2\sin^{-1}\left(\frac{a+b}{a-b}\right)$

(C) $\theta = 2\cos^{-1}\left(\frac{a+b}{a-b}\right)$ (D) $\theta = 2\sin^{-1}\left(\frac{a-b}{a+b}\right)$

***Space for rough work***

19. If 'd' is the distance between parallel tangents with positive slope to $y^2 = 4x$ and $x^2 + y^2 - 2x + 4y - 11 = 0$
(A) $10 < d < 2$ (B) $4 < d < 6$
(C) $d < 4$ (D) none of these

20. Let P be any point on a directrix of an ellipse of eccentricity e, S be the corresponding focus and C is the centre of the ellipse. The line PC meets the ellipse at A and the angle between PS and tangent at A is $\alpha$, then $\alpha$ is equal to
(A) $\tan^{-1} e$ (B) $\frac{\pi}{2}$
(C) $\tan(1 - e^2)$ (D) none of these

21. The equation $4x^2 + mxy - 3y^2 = 0$ represents a pair of real and distinct lines if
(A) $m \in R$ (B) $m \in (3, 4)$
(C) $m \in (-3, 4)$ (D) $m > 4$

22. In any triangle ABC, $\sum\left(\frac{\sin^2 A + \sin A + 1}{\sin A}\right)$ is always greater then or equal to
(A) 9 (B) 3
(C) 27 (D) none of these

23. ABCD is a square of side length a, $a \in N$, $a > 1$. Let $L_1, L_2, L_3$ ..... be points on BC such that $BL_1 = L_1L_2 = L_2L_3 = ..... = 1$ and $M_1, M_2, M_3$ ..... be points on CD such that $CM_1 = M_1M_2 = M_2M_3 = ..... = 1$, then $\sum_{n=1}^{a-1}\left(AL_n^2 + L_nM_n^2\right)$ is equal to
(A) $\frac{1}{2}a(a-1)^2$ (B) $\frac{1}{2}a(a-1)(4a-1)$
(C) $\frac{1}{2}(a-1)(2a-1)(4a-1)$ (D) none of these

24. The value of $\sum_{r=1}^{n}\frac{1}{\sqrt{a+rx}+\sqrt{a+(r-1)x}}$ is
(A) $\frac{n}{\sqrt{a}+\sqrt{a+nx}}$ (B) $\frac{\sqrt{a+nx}-\sqrt{a}}{x}$
(C) $\frac{n\left(\sqrt{a+nx}-a\right)}{x}$ (D) none of these

***Space for rough work***

25. If $\alpha$ and $\beta$ are the distinct real roots of $x^2 + ax + b = 0$ and $\alpha^4$ and $\beta^4$ are the roots of $x^2 - px + q = 0$ then the roots of $x^2 - 4bx + 2b^2 - p = 0$ are always
(A) both non-real (B) both positive
(C) both negative (D) one positive and other negative

26. The solution set of the equation $\log_x 2 \cdot \log_{2x} 2 = \log_{4x} 2$ is
(A) $\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\}$ (B) $\left\{\frac{1}{2}, 2\right\}$
(C) $\left\{\frac{1}{4}, 2^2\right\}$ (D) none of these

27. The equation of ellipse whose foci are $(\pm 5, 0)$ and one of its directrix is $5x = 36$ is
(A) $\frac{x^2}{36} + \frac{y^2}{11} = 1$ (B) $\frac{x^2}{6} + \frac{y^2}{\sqrt{11}} = 1$
(C) $\frac{x^2}{6} + \frac{y^2}{11} = 1$ (D) none of these

28. The value of x and y satisfying the equation
$\sin^7 y = |x^3 - x^2 - 9x + 9| + |x^3 - 4x - x^2 + 4| + \sec^2 2y + \cos^4 y$ are
(A) $x = 6, y = n\pi$ (B) $x = 1, y = 2n\pi + \frac{\pi}{2}$
(C) $x = 1, y = 2n\pi$ (D) none of these

29. The equation of the tangents to the conic $3x^2 - y^2 = 3$ perpendicular to the line $x + 3y = 2$ is
(A) $y = 2x \pm \sqrt{6}$ (B) $y = 6x \pm \sqrt{3}$
(C) $y = x \pm \sqrt{6}$ (D) $y = 3x \pm \sqrt{6}$

30. The domain of the function $f(x) = \log_{3+x}(x^2 - 1)$ is
(A) $(-3, -1) \cup (1, \infty)$ (B) $[-3, -1) \cup [1, \infty)$
(C) $(-3, -2) \cup (-2, -1) \cup (1, \infty)$ (D) $[-3, -2) \cup (-2, -1)$

***Space for rough work***